

Haoran Lei

Philadelphia, PA | +1 267-592-2887 | hr782357345@gmail.com

EDUCATION

University of Pennsylvania, MSE Electrical Engineering, GPA: 3.9/4.0

Philadelphia, PA

- From Sep 2025 to Dec 2026 (Expected)
- Relevant Coursework: Smart Devices, Statistics for Data Science, Graph Neural Networks, Nanorobotics, Introduction to Robotics, Feedback Control Design and Analysis, Applied Machine Learning

University of Edinburgh, BEng (Hons) Electronics and Electrical Engineering, First Class

Edinburgh, UK

- From Sep 2021 to Jun 2025
- Final Project: Implementation of a Brain-Computer Interface Model on Raspberry Pi
- Relevant Coursework: Data Analysis and Machine Learning, Digital System Design, Software and Embedded Systems, Signals and Communication Systems, Control and Instrumentation Engineering, Sensors and Instrumentation

SKILLS

Programming and Development: Python, C, C++, Verilog, Bash, Linux, Git

Robotics and Control: ROS/ROS2, MoveIt, RViz, Gazebo, URDF/Xacro/SRDF, TF, ros_control, inverse kinematics, motion planning, grasp planning, feedback control

Perception and Machine Learning: YOLOv5/YOLOv8/YOLO26, PyTorch, RGB-D perception, point-cloud processing, camera calibration, RNN, LSTM, GNN, EEG/EOG signal processing

Embedded and Hardware: Raspberry Pi 4/5, RK3568/RK3588, RKNN deployment, RealSense D435i, GPIO integration, Altium Designer, LTspice, Vivado, Xilinx SDK

LANGUAGES: Mandarin Chinese (Native), English (Fluent), Spanish (B2)

WORK EXPERIENCE

Tashan Technology

Beijing, China

Robotics Software Intern / Robotics & Embodied AI Intern

Jun 2026 – Aug 2026

- Developed an RGB-D perception pipeline for a mobile dual-arm robotic system by integrating RealSense D435i sensing, YOLO object detection, point-cloud processing, camera calibration, and coordinate-frame transformations.
- Implemented 3D object-pose estimation and integrated perception outputs with MoveIt inverse kinematics and trajectory planning to enable autonomous robotic grasping.
- Configured and validated the ROS and MoveIt software stack, including robot models, planning groups, joint limits, controllers, TF transforms, and simulation environments.
- Investigated collision-aware manipulation using Planning Scene, OctoMap, point-cloud updates, and collision objects to improve motion-planning reliability in cluttered environments.
- Diagnosed system-level failures involving camera calibration, coordinate transformations, motion planning, robot state estimation, and communication between the mobile base, dual arms, cameras, and end effectors.
- Quantized and deployed vision models on an RK3568 edge-computing platform using RKNN for onboard robotic perception.

Lighttech Engineering SL

Elx, Spain

Summer Intern

Aug 2024 – Sep 2024

- Developed and validated embedded control functions for a wireless controller used in automotive and engineering applications, including deadman safety logic, processor interfaces, display integration, charging, USB Power Delivery, and wireless communication.
- Supported hardware architecture evaluation, interface definition, prototype bring-up, and functional debugging while collaborating with mechanical and electrical engineers on design iteration.

Zhongguancun Science City City Brain Co. Ltd
Computer Vision Intern

Beijing, China
Jun 2024 – Aug 2024

- Developed and deployed eight YOLOv5/YOLOv8 computer vision models for human-action recognition and smart-city perception applications.
- Built an end-to-end computer vision workflow covering data collection, annotation, augmentation, model training, validation, deployment, and demonstration.
- Applied vision models to vehicle-color recognition, water-level estimation, tactile-paving obstruction detection, and jumping-action detection, achieving approximately 85% overall accuracy; tuned confidence thresholds and post-processing logic to reduce false detections.

PROJECT EXPERIENCE

FallGown: Inflatable Wearable Fall-Mitigation Device
(GRASP Lab, Sung Robotics Lab)

Philadelphia, PA

Master's Thesis Project and Lead Student Researcher at Penn

Nov 2025 – Present

- Lead sensing, control, and system integration for an inflatable wearable assistive robot developed in collaboration with Stanford University's CHARM Laboratory.
- Develop an IMU-based motion-processing pipeline to track body posture and distinguish normal movement from loss-of-balance conditions in real time.
- Implement closed-loop pressure-control logic that processes pressure-sensor feedback and regulates pneumatic chamber inflation to maintain target pressure.
- Develop solenoid-valve control logic that coordinates posture estimation, pressure feedback, and pneumatic actuation for real-time assistance.
- Build sensor-calibration, data-logging, parameter-tracking, and repeatable testing workflows to evaluate control stability, compliance, and system repeatability.
- Training and testing World Model in term of body posture and pressure control

Real-Time EEG/EOG Brain-Computer Interface on Embedded Hardware

Edinburgh, UK

Bachelor's Thesis Project

Sep 2024 – Jun 2025

- Developed a non-invasive brain-computer interface using EEG and EOG signals for motor-imagery classification, achieving approximately 96% validation accuracy with stable AUC performance.
- Built an end-to-end physiological-signal processing and machine-learning pipeline covering filtering, normalization, segmentation, RNN/LSTM training, evaluation, and misclassification analysis.
- Deployed real-time inference on a Raspberry Pi 5 and designed a GPIO switch and LED PCB to integrate the embedded classifier with a physical human-machine interface.

Multi-Robot Task Scheduling and Coordination with Four Franka Panda Arms

Philadelphia, PA

Course Robotics Project

Jan 2026 – Jun 2026

- Developed a mid-level task scheduler and inter-robot communication framework for coordinating four Franka Panda robotic arms in a collaborative food-assembly task.
- Implemented task-allocation and execution logic to translate customer-selected cereal and side orders into coordinated robot actions across multiple workstations.
- Enabled concurrent operation of four robotic arms by managing task dependencies, robot availability, and communication between individual robot processes.
- Integrated scheduling and communication components with the robotic manipulation pipeline to support autonomous end-to-end execution from customer order input to completed meal assembly.

PROFESSIONAL SUMMARY

Electrical Engineering MSE student at the University of Pennsylvania focused on medical and assistive robotics, embedded sensing, feedback control, and robotic system integration. Experienced in developing human-interactive robotic systems involving physiological and motion sensing, closed-loop actuation, multi-robot coordination, RGB-D perception, and embedded computing.